Efficiency of Sodium Bicarbonate Treatment for Intensive Care Unit Patients with Severe Metabolic Acidosis: BICAR-ICU Clinical Trial

Statistical Evaluation Through Stratified Tests and Multiple Comparisons

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Introduction

This study assesses the effect of sodium bicarbonate treatment in patients with severe metabolic acidosis (**pH $<$ 7.2**) on a composite **favorable** outcome combining Day-28 survival and limited organ failure (SOFA < 3). The analysis also examines how this effect varies across key clinical factors: *age*, presence of *sepsis*, and baseline renal impairment measured by the *AKIN* classification. This approach helps identify whether specific patient subgroups respond differently to bicarbonate therapy.

Materials and methods

- Stratification on three clinically relevant variables:
  - **Age**: < 65 vs ≥ 65 years
  - Renal impairment: **AKIN** 0–1 vs 2–3
  - **Sepsis** at inclusion: yes vs no
- For each variable, comparison of sodium bicarbonate vs control on the composite favourable outcome (Day-28 survival and SOFA < 3).
- Chi-square or Fisher's exact test used according to cell counts.
- Six tests at $\alpha = 0.05$ increase the risk of at least one type I error to $1 - (1 - 0.05)^6$, so multiple-testing corrections were applied to the p-values :
  - Bonferroni and Holm (control of family-wise error rate)
    Bonferroni : $pv\_adj_i = pv_i \times m$
    Holm, m pv ranked ascending :
    $pv\_adj_i = min(pv_i(m + 1 - i), 1)$
  - Benjamini–Hochberg (control of false discovery rate) by ranking pv ranked ascending :
    $pv\_adj_i = min(pv_i \times \frac{m}{i}, 1)$
- Interaction analyses: age $\times$ sepsis, age $\times$ AKIN, and combined models when subgroup sizes were sufficient.

Results

The table below presents the raw and adjusted p-values for each stratification variable, based on chi-square or Fisher tests assessing the effect of sodium bicarbonate on the composite favourable outcome.

Table 1: Adjusted p-values across stratification variables

Strat	p	Bonf	Holm	BH
Sep = 0	0.031	0.191	0.127	0.063
Sep = 1	0.115	0.692	0.346	0.173
Age < 65	0.005	0.029	0.030	0.026
Age ≥ 65	0.235	1.000	0.471	0.283
AKIN 0–1	0.323	1.000	0.471	0.323
AKIN 2–3	0.008	0.051	0.042	0.025

Patients younger than 65 and those in AKIN stage 2–3 showed a statistically significant association between sodium bicarbonate and a favourable outcome.

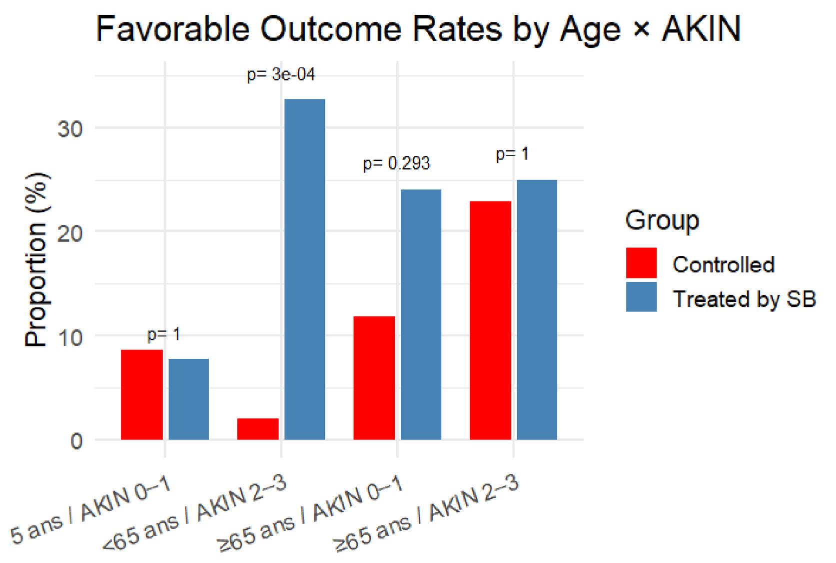

Figure 1: Age × AKIN

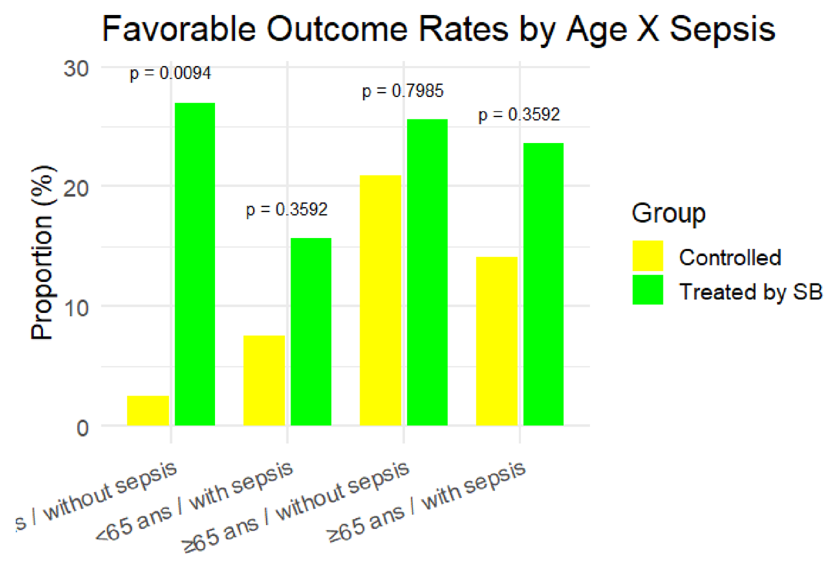

Figure 2: Age × sepsis

The subgroup of patients younger than 65 years without sepsis showed a marked response to treatment, with a p-value of 0.009. The treatment effect was also pronounced among those classified as AKIN 2–3.Across all subgroups, the proportion of patients achieving a favorable outcome was consistently higher among those who received sodium bicarbonate compared with those who did not.

We could not perform a test with all the three variables combined, because sample sizes became very small.

Conclusions

The BICAR-ICU trial evaluates whether sodium bicarbonate improves outcomes in ICU patients with severe metabolic acidosis. Our analyses suggest that bicarbonate therapy is associated with a higher probability of a favorable composite outcome, particularly in patients younger than 65 years and those with AKIN stage 2–3. Further work could extend these results using more flexible models and additional covariates.

GitHub

All analyses and code are available at: https://github.com/ksarih/Data-In-Lab

Literature cited

The analysis relied on the BICAR-ICU trial dataset and in addition to the following documents :

- Giraud C. Tests multiples en statistique. HAL, 2022.
- NIST. Engineering Statistics Handbook – Chi-Square Tests.

Acknowledgements

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